

TPL Signum - RF Intruder Detection System

SIGFNE4TST

Demo Kit - 2.4Ghz 20dB 1-Ellipse 2-Hops USB-CLI Configuration Manual

1. Technology Introduction

Our company, TechPack Lab, abbreviated as TPL, has invented and developed a unique intruder detection technology and system, called Signum. The Signum technology is based on RF, but unlike RF technologies like radar, our system doesn't have a directivity, and detection units can be located anywhere and in any orientation. Furthermore the units are extremely small, extremely low-power, and very low-cost.

The Signum technology is deployed together with another unique technology of ours, the TPL-Mesh. TPL-Mesh is an extremely optimized Mesh wireless communication. It allows participating units, like the Signum units, to be active and relaying in mesh for years in minimal batteries. The TPL-Mesh units can come in various throughput levels, from highest HD video-grade level to lowest levels as incorporated in miniature Signum units. While incorporated into Signum units it efficiently shares the same processor and transceiver, thus allowing extremely efficient sizes (down to 1.5x1.5cm PCBs), extremely low power (well under 1mA) and very accessible pricing for very high quantity deployments.

The Signum units come in multiple form factors, operating frequencies and management methods. Coupled with TechPack Lab's design services capabilities, these units can be easily adapted to any required use case.

2. SIGFNE4TST Demo Kit

The current SIGFNE4TST demo kit is meant to function as an early demonstration kit, showing both the detection capabilities together with Mesh communication capabilities. This demo kit is working in standard globally certified unlicensed 2.4GHz band, and are transmitting at the maximally certification-allowed 20dBm RF transmission power at this frequency.

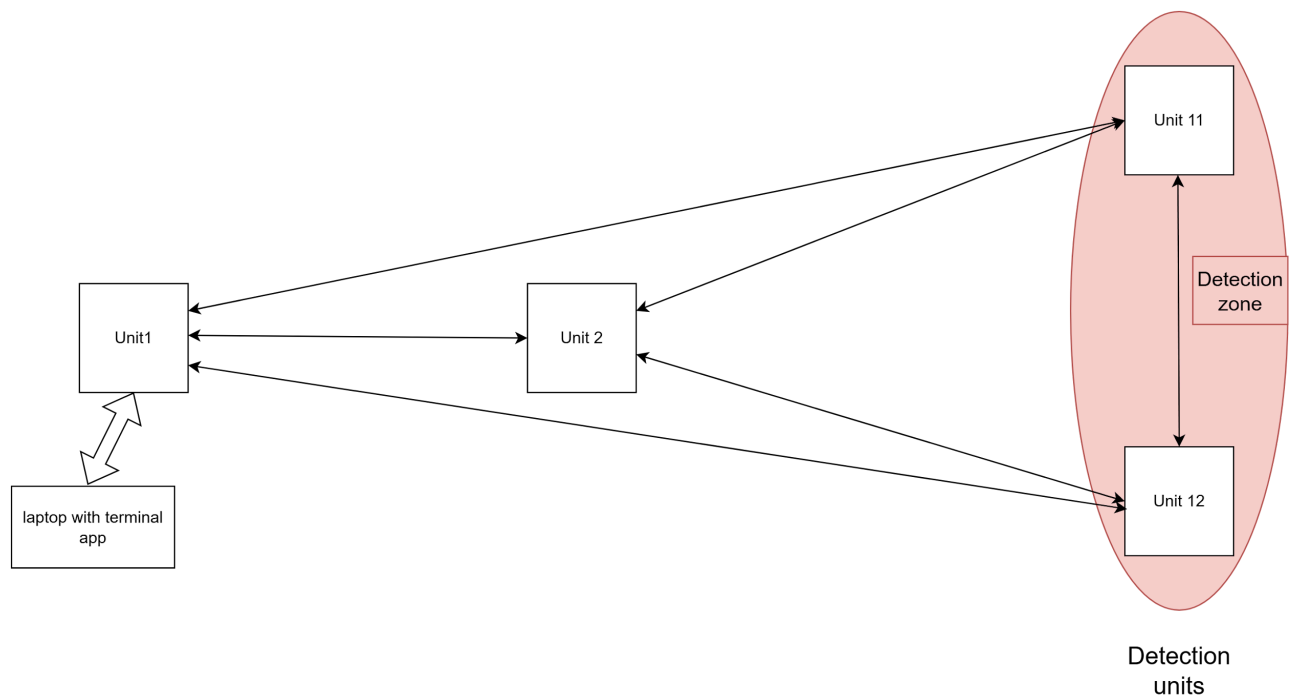
The SIGFNE4TST demo kit consists of 4 units:

- 2 detection units - creating one detection Ellipse area (they also function in a mesh, relaying data through each other)
- 1 Control unit which connects over USB to a control PC

- 1 Mesh-Relay unit to allow further extending the distance between the Control Unit and the detection units, making it a 2-hops Mesh network.

TPL-Signum units are numbered 1,2,11 and 12. Units 11 and 12 are detection units. Unit 1 is connected to a terminal through the USB port. Unit 2 is placed between unit 1 and the detection units. Intruder detection events which are detected between 11 and 12 are transmitted directly to unit 1 (if in range) or through unit 2 (when 1 is not in direct range). Unit 1 displays one or more lines on the terminal indicating that the event has occurred:

The following is a diagram of the system:



3. Installation in the field

Units are packaged in a basic container to withstand some environmental conditions. They are powered by a battery pack containing 2 AA batteries. To power on and off, the top of the container should be removed by unscrewing four screws, and the switch on the battery pack can be turned on or off.

The units can be turned on in any order and will connect and start working.

The units are communicating in full mesh, so if unit 1 is not in range of units 11 or 12, it can still report on detection through unit 2, if unit 2 is within range of either units 11 or 12, or both.

If after the units are already working, they are moved, then the system will adapt to the new structure. For example, if unit 2 was installed to be within range of 11, and after a long time it's moved to be in range with 12, then the system will adapt automatically and continue working properly.

If unit 1 is powered-on and eventually there are no proper connections to units 11 and 12, then the system will obviously not work. Unit 1 will print a message indicating that a disconnection has occurred.

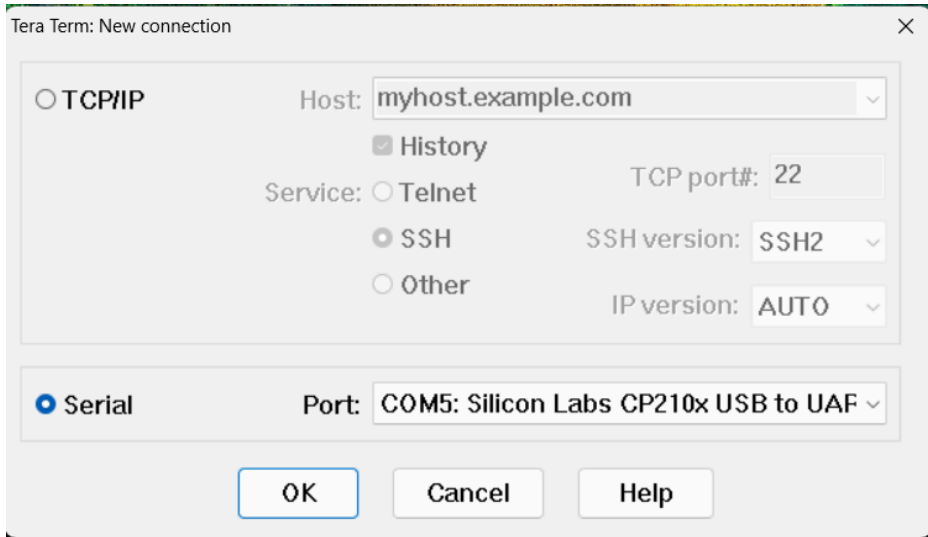
It is recommended that after the installation completion a proper detection trigger is tested in order to close the full loop of correct operation. It is also recommended to check the set after installation by issuing a map command on Unit 1, to make sure that everything is working fine, all units are in place, and average RSSI values are sufficient for stable operation, and all Gain/Threshold are configured to the right values.

Configuration commands can be issued through the terminal on unit 1. After issuing configuration commands it is highly recommended to verify that update settings have reached all the units using the **map** command.

Working with the system requires a simple serial terminal tool installed on a PC.

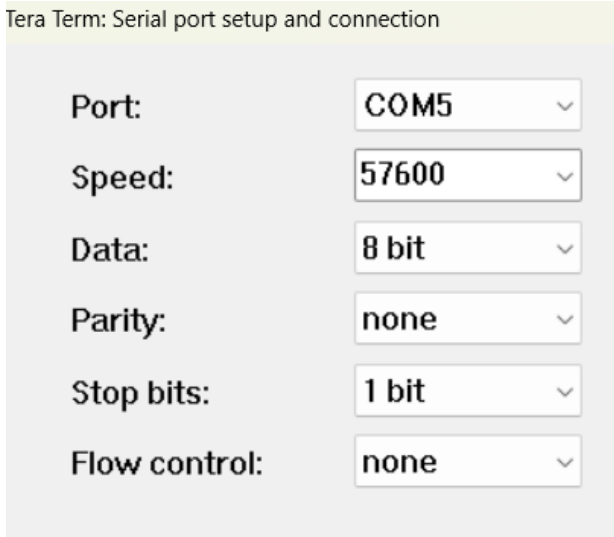
A good tool example on Windows is TeraTerm:

<https://teratermproject.github.io/index-en.html>



The screenshot shows the 'Tera Term: New connection' dialog box. It has two main sections. The top section is for TCP/IP connections, with 'Host' set to 'myhost.example.com', 'History' checked, 'Service' set to 'SSH', 'TCP port#' set to '22', 'SSH version' set to 'SSH2', and 'IP version' set to 'AUTO'. The bottom section is for Serial connections, with 'Serial' selected and 'Port' set to 'COM5: Silicon Labs CP210x USB to UAF'. At the bottom are 'OK', 'Cancel', and 'Help' buttons.

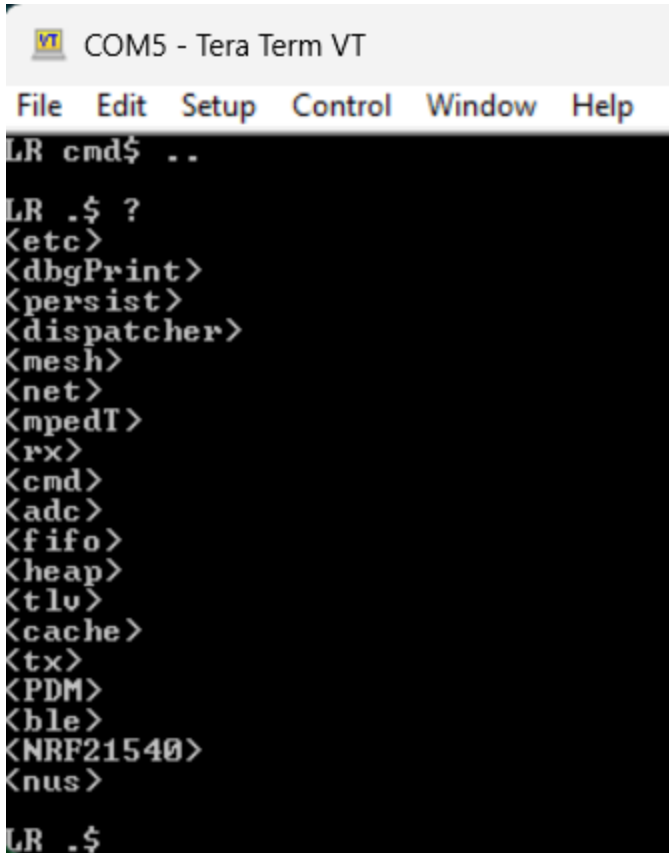
Configuration for serial connection in TeraTerm:



The screenshot shows the 'Tera Term: Serial port setup and connection' dialog box. It contains several settings, each with a dropdown menu: 'Port' is set to 'COM5', 'Speed' is set to '57600', 'Data' is set to '8 bit', 'Parity' is set to 'none', 'Stop bits' is set to '1 bit', and 'Flow control' is set to 'none'.

The TeraTerm tool has multiple useful features which can be used, like logging with Timestamp and Scripting. Scripting capabilities can trigger any other operation through `exec()` command which can trigger Python and other tools to allow event reporting over internet and more.

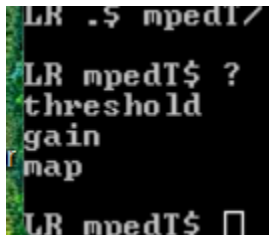
To start working need to enter the `mpedT` folder by typing it with Enter or starting to type with “mp” and using tab key to complete the command:



```

VT COM5 - Tera Term VT
File Edit Setup Control Window Help
LR cmd$ ..
LR .$ ?
<etc>
<dbgPrint>
<persist>
<dispatcher>
<mesh>
<net>
<mpedI>
<rx>
<cmd>
<adc>
<fifo>
<heap>
<tlv>
<cache>
<tx>
<PDM>
<ble>
<NRF21540>
<nus>
LR .$
  
```

Typing “?” with enter will give a list of commands or folders.



```

LR .$ mpedI/
LR mpedI$ ?
threshold
gain
map
LR mpedI$ 
  
```

Key UP is history, which also supports completion, like in Linux.

4. Intruder Detection Events

The following is an example of detection events which can be seen on the terminal:

```

[1862125] I CMD:DETECTION ece0(11)-a644(12) th:500 val:555 c:1
[1865126] I CMD:DETECTION ece0(11)-a644(12) th:500 val:542 c:1
[1865201] I CMD:DETECTION a644(12)-ece0(11) th:500 val:551 c:2
[1865726] I CMD:DETECTION ece0(11)-a644(12) th:500 val:709 c:1
[1866746] I CMD:DETECTION ece0(11)-a644(12) th:500 val:635 c:3
  
```

The most important value in this line is its existence. If such a line is printed, it means that sufficient movement has been detected, and an alert has been triggered.

The *val* field has a significance of indicating a kind of level of detection. It doesn't directly indicate any physical meaning, but in general the higher the value, the higher chance that a real movement has happened.

Detection based on communication loss:

The DETECTION-COMM alert indicates that there has been some obstruction between detection units which prevents communication between them. Also if one of the detection units suddenly dies (or gets powered off) this will be the alert.

```
[58799] I CMD:DETECTION-COMM A644(12)-ECE0(11) 2025  
[58800] I CMD:DETECTION-COMM ECE0(11)-A644(12) 2041
```

Display of detection messages to the terminal can be stopped if, for example, they are being displayed so frequently that it is not possible to execute any CLI commands. To stop display of detection messages, enter the *cmd* folder and execute *re* without any parameters.

```
LR .$ cmd  
LR cmd$ re  
LR cmd$
```

To start display of detection messages, enter *re 3 4*:

```
LR cmd$ re 3 4
```

5. Connection/disconnection events

If a connection or disconnection occurs between one or more devices, a message will be printed to the terminal.

The following is an example of a disconnection message, where unit 2 and unit 11 have lost connection. Disconnection messages often follow a series of DETECTION-COMM events. The 0 at the end of the line indicates disconnection and in following case unit 2 reports that it got disconnected from unit 11:

```
[1531538] I CMD:CONNECTED 376F(2) connected:ECE0(11) 0
```

The following is an example of a message of connection between unit 1 and unit 2. The 1 at the end of the line indicates connection:

```
[1531660] I CMD:CONNECTED EDC5(1) connected:376F(2) 1
```

6. Sensitivity Configuration Commands

It's possible to configure some parameters of the current detection algorithm. These parameters will be saved in non-volatile memory, so it will be used in all upcoming runs, including through battery replacements.

The terminal has a menu system which allows performing multiple operations on the units. The user configuration for this system is located under the mpedT subfolder.

In order to see the list of available folder or commands, you need to type "?" and press Enter.

The following will appear:

```
.$ ?  
<etc>  
.  
.  
<mpedT>
```

In order to get into mpedT folder of commands, it's needed to type in "mpedT" and press Enter. (an alternative is to press "mp", then press Tab for autocompletion, and then press Enter).

The following prompt will appear, indicating that you are now in the correct folder in the menu:

```
mpedT$
```

If you are not able to see or enter into the mpedT menu, then you are probably in a different folder. Type / with enter to get into root folder. Type .. with enter to get into high folders.

To see the available commands in the folder, it's needed to issue "?" command again, to get the following options:

```
mpedT$ ?  
threshold  
gain  
map
```

The current threshold and gain commands allow configuring different parameters for the detection sensitivity.

a. Threshold command

threshold command takes one value as a parameter.

The threshold parameter directly configures the threshold of the last “val” parameter in alert printout, above which the alert will happen and will be printed on the screen.

Example:

LR mpedT\$ threshold 250

[694678] I CMD:th3:250

Decreasing threshold will increase the sensitivity, and thus the frequency of printouts if many of them exist, while increasing threshold will have the opposite effect.

In order to better understand the current environment and detection capabilities it's possible to significantly decrease the threshold parameter, which will result in many prints, allowing the user to learn the system behavior. After learning is completed, and other parameters (i.e. Gain) are configured, the threshold could be set to a sufficiently high level, so that an existence of printout will indicate a real detection event and will prevent false-alarms.

After issuing the Threshold configuration, remember to verify that it was properly received by all units using the map command.

b. Gain command

Gain command receives a single parameter and configures the strength of the internal filter.

With higher gain values the filtering will be weaker, alerts will be raised closer to the time of the event, but there is also a higher chance for noise and alerts without an actual movement.

Again, it's suggested to learn the effects of Gain setting when threshold is configured sufficiently low to see many printed alerts, and in such case Gain can be lowered for stronger filtering which will prevent single high detection events, but could also filter out real movement and add a small delay to the detections.

Example:

mpedT\$ gain 16

Value of 16 is a value that in some cases has been found to be sufficiently strong for outdoor operation and be resistant to some effects like basic wind.

After issuing the Gain configuration, remember to verify that it was properly received by all units using the map command.

7. Retrieving system status - Map command

Map command prints information about remote devices configuration and connections between devices.

The commands structure shows which units are in range and connected to which units, including the average RSSI between them. The max RSSI after which the communication stops is roughly about -90dB.

Additionally, it's possible to read the gain and threshold settings in the set, and it's important to verify that all units in the set have the correct gain and threshold settings configured.

It's also possible to learn the internal identifiers of the units this way to help identify units when multiple sets are in the range, or when stickers fall, etc.

The structure of the commands with interesting parameters explained:

[<timestamp, milliseconds from last powerup/reset, with a wraparound at approximately 49 days>] I **CMD:MAP_RSP** from <identifier of unit which sent this response> **gain:**<gain setting> **voltage:**<reading of battery, when supported> **scan:1 adv1:** [<identifier to first unit connected to the current unit> **th3:**<detection threshold configuration between current unit and this one, and 0 when disabled, that is expected in non-detection connections 1-2,1-11,1-12,2-11,2-12> <average RSSI of communication between these units>**dBm dt:240**] [<Similar structure for second device possibly connected to this unit>] [<Similar structure for third device possibly connected to this unit>]

Example:

LR cmd\$ map

[5044419] I CMD:localidx:1

[5044419] I CMD:idx:1 idx:1

LR cmd\$

[5044913] I CMD:MAP_RSP from 2 ver:0.4c10 gain:64 voltage:2908 scan:1 adv:1: [12 th3:0 -57dBm dt:180] [11 th3:0 -36dBm dt:240] [1 th3:0 -71dBm dt:300]

[5045162] I CMD:MAP_RSP from 11 ver:0.4c10 gain:64 voltage:2925 scan:1 adv:1: [1 th3:0 -78dBm dt:421] [12 th3:500 -52dBm dt:300] [2 th3:0 -39dBm dt:241]

[5045529] I CMD:MAP_RSP from 12 ver:0.4c10 gain:64 voltage:3035 scan:1 adv:1: [11 th3:500 -50dBm dt:300] [2 th3:0 -58dBm dt:180]

We can learn from there the following:

This command was issued from unit 1, and 3 units responded:

- Unit 2 is connected to units 1, 11, and 12. RSSI threshold is 0 between Unit 2 and all other units connected to it, as no detection takes place on Unit 2. Unit 11, with RSSI -36 dbM, is closer than unit 12 with RSSI -57 dbm. Both units are sufficiently close for very stable communication.
- Unit 11 is connected to units 1, 2 and 12. The threshold between unit 11 and unit 12 is 300, as these are the detection units.
- Unit 12 is connected to units 11 and 2. Note that unit 12 is not connected directly to Unit 1, but communicates through units 11 and 2 through the BLE mesh.
- Gain of all units is properly configured to 64.

8.DFU - Firmware update

The update is performed using nRF Connect for Mobile on Android or iOS.

nRF Connect for Mobile

1. Entering DFU mode:

To update units other than unit 1, in TPL_CLI cmd menu execute:

```
cmd$ enterBootloader -i xx
```

where xx is the Unit number of the board to be updated (2,11, or 21) After the command is executed, board xx will enter DFU mode.

To update unit 1, exit the cmd folder and enter the *etc* folder, then execute:

LR etc\$ enterBootLoader

After pressing enter , the following will be printed to the terminal:

ENTERING BOOTLOADER

After the command is executed the board will enter DFU mode, so remote commands to the device (like map) and mesh communication through it will not work, and location connection to the terminal (if attached) won't work either.

2. Connecting over BLE

Run nRF Connect for mobile phone. Enter dfu as filter

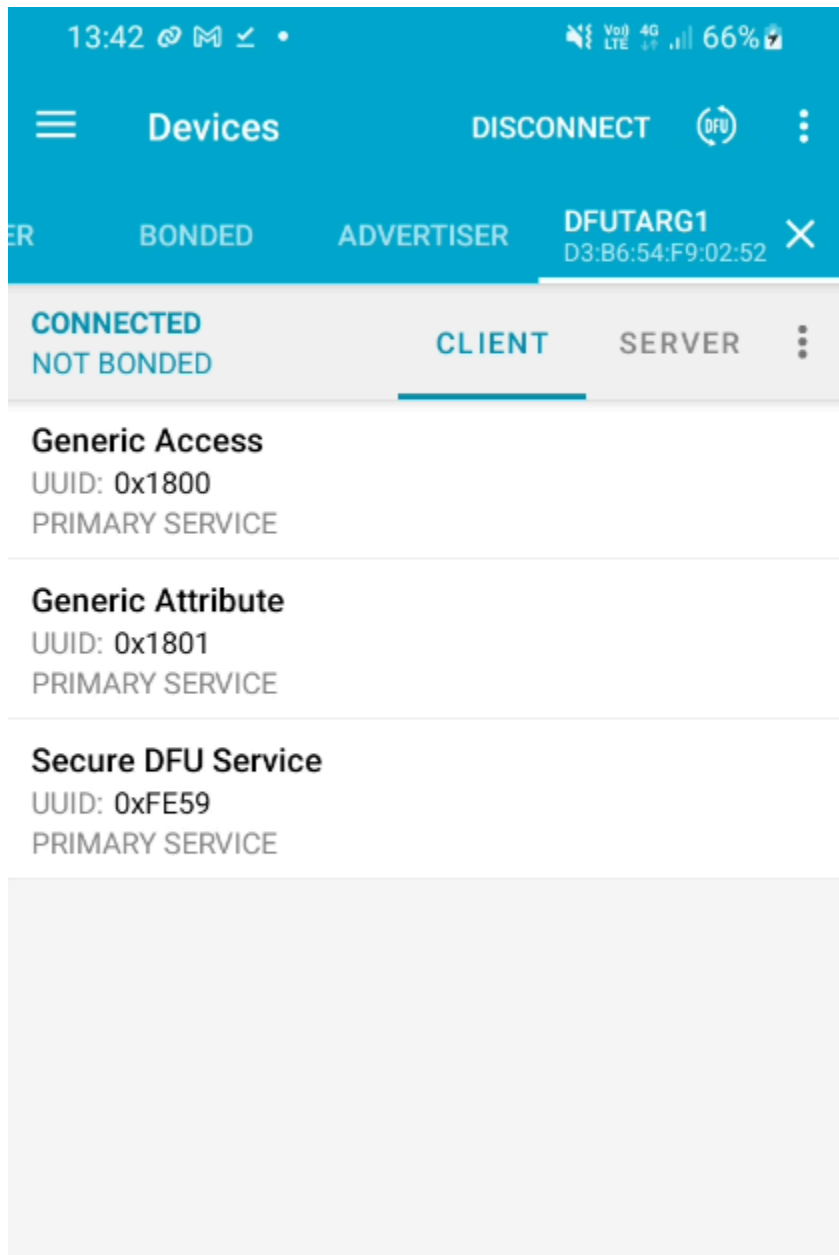
Start scan. Bootloader will show up as device DfuTarg1

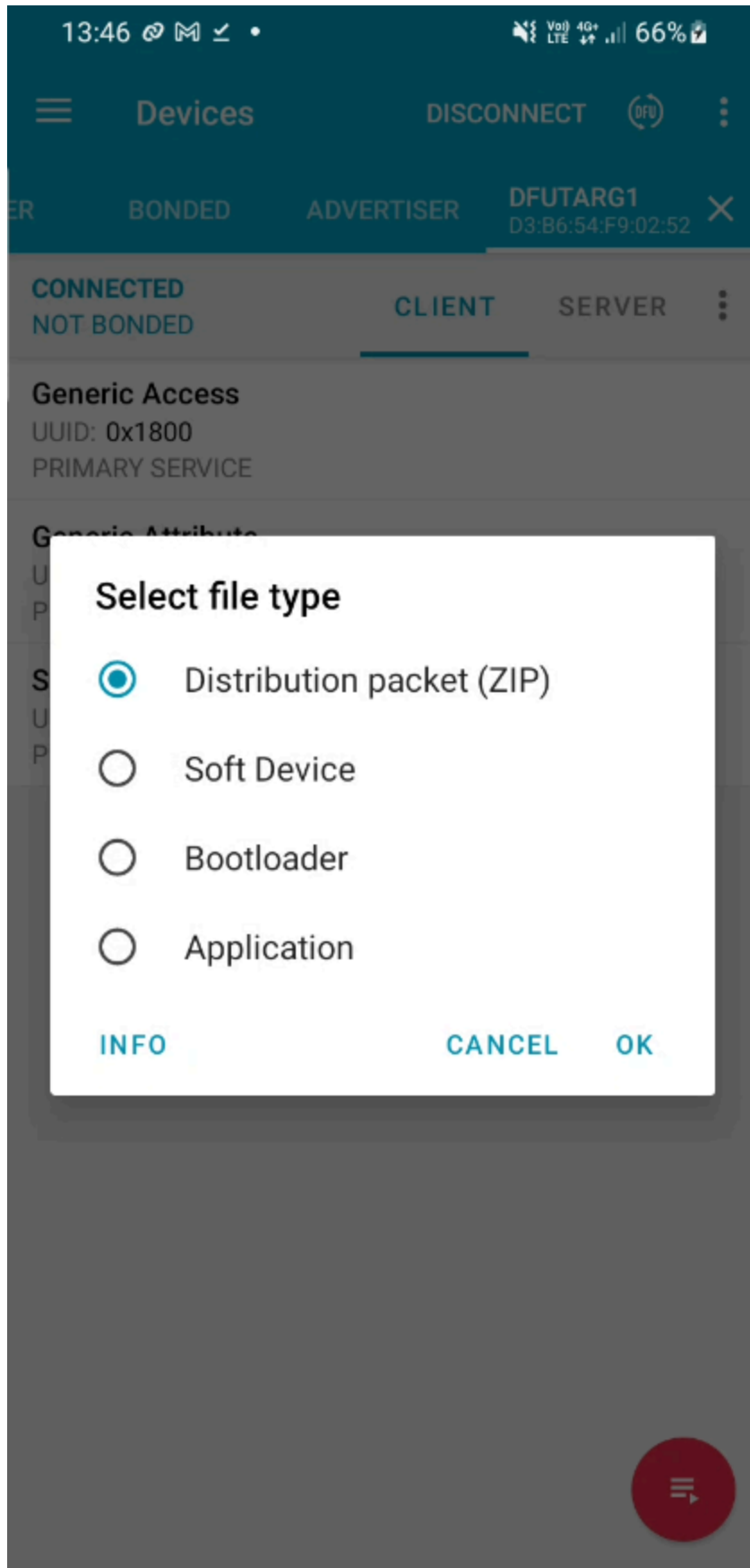
Connect to DfuTarg1 :

If connection is not made within 2 minutes after entering bootloader mode, the bootloader will exit bootloader mode and load the existing application (process will need to be restarted to re-enter bootloader again).

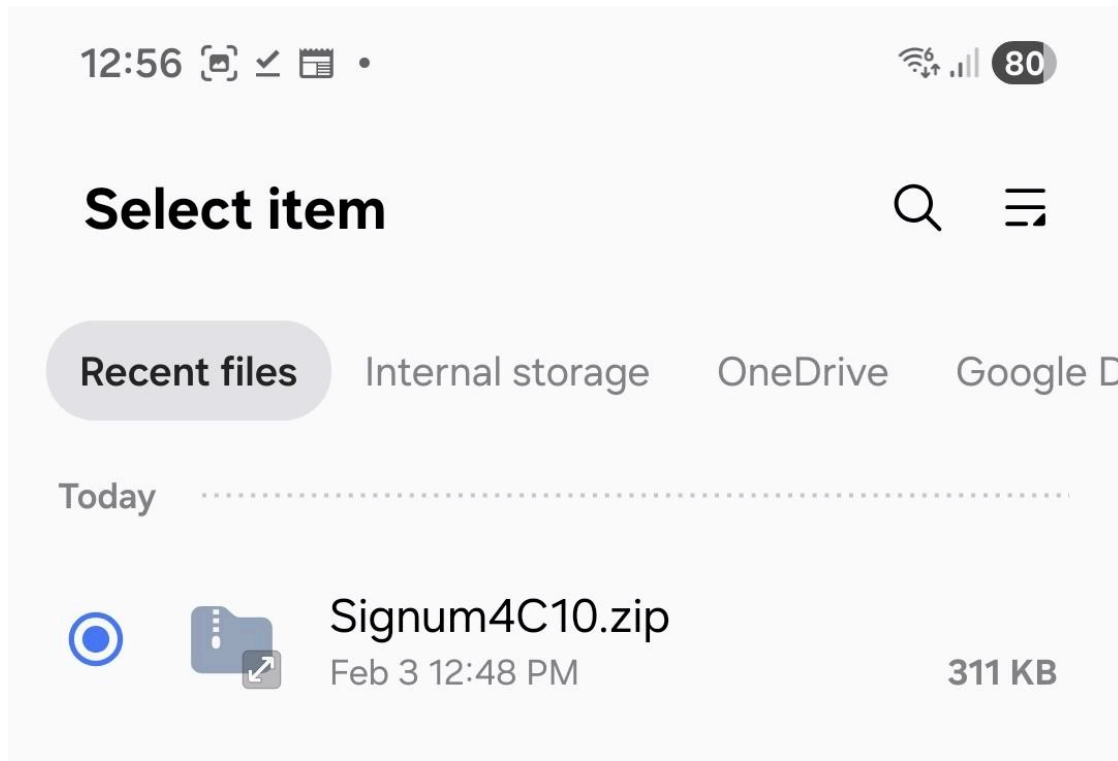
3. Transferring application file

On the DfuTarg1, the DFU icon (to the right of DISCONNECT) will start secure DFU. Click the icon and a window for choosing zip file to transfer will appear:

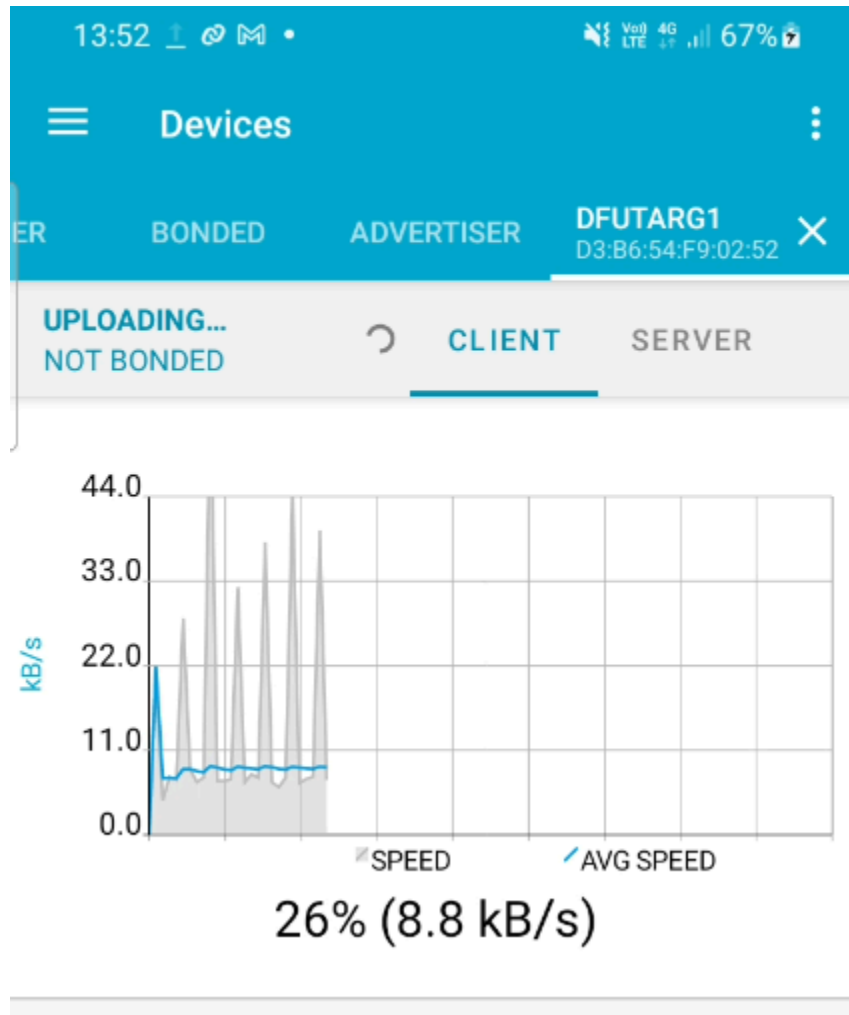




Choose .zip file The file name will change depending on the board type and version number:



Click on **Start DFU** and transfer will start.



The transfer takes 1-2 minutes. On completion, the updated application has been loaded and the board will start advertising as mesh xx

The updated application has been loaded and should be running.

Notes